AI-Integrated Multi-Modal Academic Evaluation, Performance Intelligence, and Automated Viva Assessment System

R26-SE-025

Project Proposal Report

Wickramarathne H.I.B – IT22239198

B.Sc. (Hons) Degree in Information Technology Specialized in Software Engineering

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

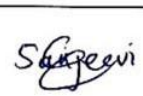
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Wickramarathne H.I.B	IT22239198	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Ms. Sanjeevi Chandrasiri	14/3/2026	

ABSTRACT

The evaluation of student examinations is traditionally performed through manual grading of handwritten answer scripts. This process can lead to inconsistencies due to factors such as examiner bias, fatigue, and subjective interpretation of responses. In large academic environments, manual grading also requires significant time and effort, resulting in delays in providing feedback to students. Although several Artificial Intelligence (AI) based grading systems have been proposed, many of these approaches require students to submit typed responses and are typically designed for limited question formats. However, in many universities, including our own institution, examinations are still conducted using handwritten answer papers containing textual responses such as essays, short answers, and explanatory answers. This creates a gap between existing automated grading technologies and real world examination practices.

This research proposes an AI based automated exam evaluation framework capable of processing both handwritten answer scripts and computerized or online answers while analyzing multiple textual question formats. The system includes a web or mobile based interface that allows examiners to upload images of physical answer papers. These images are processed using Optical Character Recognition (OCR) and image preprocessing techniques to extract machine-readable text. The extracted responses are then analyzed using Natural Language Processing (NLP) techniques and transformer based language models to interpret student answers, evaluate conceptual understanding, and assign appropriate grades. The proposed system aims to improve grading consistency, reduce evaluation time, and enable reasoning-based assessment rather than simple keyword matching. Additionally, the framework generates explainable feedback to provide transparency in grading decisions.

The proposed approach has potential social and commercial impact by supporting scalable and fair examination evaluation in educational institutions and enabling integration with modern digital learning and assessment platforms.

Keywords: *Automated Answer Script Evaluation, Natural Language Processing in Education, AI based Exam Grading, Handwritten Text Recognition, Explainable AI in Assessment.*

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my Supervisor and Co-Supervisor for their continuous guidance, valuable suggestions, and encouragement throughout the development of this research project. Their academic support and constructive feedback greatly contributed to improving the quality and direction of this study. I would also like to thank my team members for their cooperation, dedication, and collaborative effort during the project. Their support in discussions, research activities, and system development played an important role in the successful preparation of this proposal.

My sincere appreciation is extended to the academic and administrative staff of the Sri Lanka Institute of Information Technology (SLIIT), particularly the Department of Information Technology, for providing the knowledge, learning environment, and resources necessary to carry out this research. Finally, I am grateful to everyone who directly or indirectly supported me in completing this project proposal.

TABLE OF CONTENT (Auto Generated)

DECLARATION 2
ABSTRACT 3
ACKNOWLEDGEMENT 4
TABLE OF CONTENT (Auto Generated) 5
LIST OF TABLES 6
LIST OF FIGURES 7
LIST OF ABBREVIATIONS 8
LIST OF APPENDICES 9
1. INTRODUCTION 10
1.1 Background and Context 10
1.2 Literature Review 11
1.2.1 Existing Research Approaches 11
1.2.2 Existing Products and Services 11
1.2.3 Limitations of Existing Systems 12
1.3 Research Gap 12
1.3.1 Incompatibility with Handwritten Examination Workflows 12
1.3.2 Fragmented Support for Diverse Question Formats 13
1.3.3 Overreliance on Surface Level Keyword and Similarity Metrics 13
1.3.4 The "Black Box" Problem - Lack of Explainable Feedback 13
1.3.5 Disconnect from Real World Examination Environments 13
Justification of Novelty and Creativity 14
Evidence Based Argument 14
2. OBJECTIVE 15
2.1 Main Objective 15
2.1.1 SMART Analysis of the Main Objective 15
2.2 Specific Objectives 16
3. METHODOLOGY 18
3.1 Research Approach 18
3.2 Technical Methods and Algorithms 18
3.3 Tools and Platforms 19
3.4 Data Collection and Ethical Considerations 19
3.5 Validation Strategy and Performance Metrics 20
4. HIGH-LEVEL SYSTEM ARCHITECTURE 21
5. USER REQUIREMENTS 22
5.1 Stakeholder Identification 22
5.2 Functional Requirements (FR) 22
5.3 Non Functional Requirements (NFR) 23
6. COMMERCIALIZATION PLAN 24
7. BUDGET AND JUSTIFICATION 25
8. WORK BREAKDOWN STRUCTURE (WBS) 26
8.1 Task breakdown 26
8.2 Timeline Alignment 27
8.3 Balanced workload distribution 28
9. GANTT CHART 29
10. REFERENCES LIST 30
11. APPENDICES 31

LIST OF TABLES

Table 1: List of Abbreviations 8
Table 2: Budget justification 25
Table 3: Timeline Alignment 27
Table 4: Balanced workload distribution 28

LIST OF FIGURES

Figure 1: High level system diagram 21
Figure 2: Work Breakdown Structure 26
Figure 3: Gantt chart 29

LIST OF ABBREVIATIONS

Table 1: List of Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
ASAG	Automatic Short Answer Grading
CoT	Chain of Thought
DSR	Design Science Research
FR	Functional Requirement
GDPR	General Data Protection Regulation
GNN	Graph Neural Network
HITL	Human in the Loop
IRR	Inter-Rater Reliability
LIME	Local Interpretable Model-agnostic Explanations
LLM	Large Language Model
LMS	Learning Management System
MOOC	Massive Open Online Course
NFR	Non-Functional Requirement
NLP	Natural Language Processing
OCR	Optical Character Recognition
PII	Personally Identifiable Information
SaaS	Software as a Service
SHAP	SHapley Additive Explanations
SMART	Specific, Measurable, Achievable, Relevant, and Time-bound
UI/UX	User Interface / User Experience
WBS	Work Breakdown Structure
WCAG	Web Content Accessibility Guidelines
XAI	Explainable Artificial Intelligence

LIST OF APPENDICES

1. INTRODUCTION

1.1 Background and Context

The evaluation of students' examinations is an essential part of how educational institutions determine what students have learned. Most often, teachers' grades are based on students' answers that they write on pieces of paper. This approach gives teachers an opportunity to understand students' responses and be flexible in evaluating students' work. Despite this, there are many challenges associated with this traditional approach. For example, it takes a lot of time to grade more students' answer scripts, especially if there are many students in class. Furthermore, teachers are likely to be exhausted and fatigued after spending more time grading students' work.

There are various problems in the sector. For instance, people in the sector face various challenges. For example, students in the sector have to wait for a certain period of time before receiving their grades. This affects how they learn. Secondly, it takes teachers in the sector a lot of time grading students' examinations. This affects how they work. This may also cause errors in the grades of the students in the future. There is also a challenge in managing grades in the institutions in the educational sector. There are various problems in the educational sector. For example, students and teachers in the sector face various challenges. For instance, there is a challenge in managing grades in the sector. In recent years, advancements in Artificial Intelligence (AI) and Natural Language Processing (NLP) have led to the development of automated grading systems that can assist in evaluating written responses. Some of these systems are capable of grading short answers and essays, helping to reduce grading time and improve efficiency. However, many existing systems work only with typed answers and are designed for limited types of questions. In many universities, including those here in Sri Lanka, students still write their exam answers by hand. Because of this, many automated grading systems cannot be easily applied in real examination environments.

To overcome this problem, the research is proposing the use of Artificial Intelligence for evaluating both handwritten and typed answers. The proposed system will include a web-based or mobile application through which teachers can upload images of answer scripts. Optical Character Recognition (OCR) technology will be used for recognizing the images and converting the answer scripts into text format. After that, Natural Language Processing (NLP) will be used for analyzing the answer scripts and providing appropriate grades along with the reasons for the grades provided.

This research is also in line with the United Nations' Sustainable Development Goal 4, which is concerned with improving the quality and accessibility of education. By improving the efficiency of the grading system, this system can be helpful in improving the quality of assessment in educational institutions. The research is also in line with research areas in the institution concerning Artificial Intelligence and educational technology.

Developing systems that can assist with exam evaluation is an important step toward improving modern education systems. Such technologies can help institutions manage large numbers of students while maintaining fair and consistent grading practices.

1.2 Literature Review

The evolution of automated grading systems has transitioned from simple pattern matching to complex neural architectures. This review examines 8 scholarly sources to discuss existing approaches, products, and the persistent limitations that necessitate a unified AI evaluation engine.

1.2.1 Existing Research Approaches

Semantic and Ontology Based Analysis: Early advancements moved beyond keyword matching to holistic evaluation. Rokade et al. (2018) proposed a system using Natural Language Processing (NLP) and ontology to extract synonyms and domain related words, allowing for "holistic" evaluation that considers word combinations and conceptual coverage. [1]

Transformer and LLM Architectures: Recent research leverages deep learning. Gobrecht et al. (2024) introduced an Automatic Short Answer Grading (ASAG) system based on fine tuned open source transformer models. Their study demonstrated that these models could achieve higher consistency than human experts, with a median absolute error 44% smaller than that of human re graders. [2]

Graph Based and Hybrid Techniques: Suresh et al. (2023) developed an automated essay grading system that combines NLP with graph based techniques to analyze syntax, semantics, and grammar. [3]

Mixed Methods and Perception Studies: Ragolane and Patel (2024) employed an explanatory sequential mixed methods design to investigate student perceptions, revealing that while students value the speed and consistency of AI, they remain skeptical without human oversight. [4]

1.2.2 Existing Products and Services

Scantron Systems: A foundational technology in automated assessment, the Scantron has long been used for grading multiple choice tests efficiently, though it is limited to objective formats. [5]

Exam Trainers and Practice Apps: IU International University of Applied Sciences utilizes an "Exam Trainer" feature where AI models grade student answers during practice sessions to predict exam performance and award certifications. [2]

Generative AI Platforms: Tools like ChatGPT are increasingly used in higher education for generating ideas for application-based questions and providing immediate, though sometimes unverified, feedback on theory based answers. [4]

Massive Open Online Courses (MOOCs): Platforms like Coursera or edX utilize various automated grading techniques to manage the massive influx of student assignments, though they often struggle with complex programming or open ended tasks. [5]

1.2.3 Limitations of Existing Systems

Lack of Multimodal Integration: Most systems focus on a single format either text based essays (ASAG) or programming code. There is a significant gap in systems that can cross verify knowledge across text, diagrams, and oral examinations (vivas). [5]

Algorithmic Bias: Najafi (2024) found that AI grading systems can exhibit significant bias; for example, non native English speakers received 8% lower average grades from AI than native speakers, highlighting a failure to focus on core conceptual understanding over linguistic perfection. [6]

The "Black Box" and Trust Deficit: Wang (2025) and Ragolane (2024) both highlight that existing AI models lack "human touch" and transparency. Students express concern over the lack of explainable feedback and the potential for increased stress due to the rigor of AI focusing on precise "correctness" over general understanding. [4]

UI/UX and Accessibility Barriers: A literature review by Aldriye (2019) noted that many automated systems suffer from poor user experience, beginner syntax errors, and language barriers, making them difficult for students to use effectively. [5]

Ethical and Integrity Concerns: The use of generative AI in grading introduces risks of "human unintelligence" and unethical use if not paired with robust policy frameworks and human in the loop validation. [4]

1.3 Research Gap

The analysis of current scholarly literature and existing technological frameworks reveals five critical deficiencies in the landscape of automated academic evaluation. Despite the progress in Natural Language Processing (NLP), a significant gap remains between theoretical research and the practical needs of traditional university environments.

1.3.1 Incompatibility with Handwritten Examination Workflows

What is missing: Most automated grading systems are designed exclusively for typed responses submitted via digital platforms and lack a system to analyze and grade physical papers or scripts. [7]

Why current approaches are insufficient: In many global academic contexts, including universities in Sri Lanka, examinations are still conducted using physical, handwritten answer scripts. Existing research models are often validated using "clean" digital datasets, which fail to represent the noise, varied handwriting styles, and structural complexities of real world paper based assessments. [7] [2]

Research Gap: There is a distinct lack of integrated systems capable of bridging the physical digital divide. This research addresses this by combining handwritten script digitizing, OCR based text extraction, and AI based evaluation into a continuous workflow. [5]

1.3.2 Fragmented Support for Diverse Question Formats

What is missing: A majority of current models, such as those discussed by Gobrecht et al. (2024), are optimized for a single type of assessment, most notably Automatic Short Answer Grading (ASAG). [2]

Why current approaches are insufficient: Real world examinations are multi format, they consist of a mix of short answers, long form essays, and structured explanations. Most current models cannot handle these varying structural and length requirements within a unified system, requiring instructors to use multiple tools. [2]

Research Gap: There is a critical need for a unified evaluation framework capable of analyzing multiple textual question formats concurrently while maintaining grading accuracy and structural integrity across different answer lengths. [2]

1.3.3 Overreliance on Surface Level Keyword and Similarity Metrics

What is missing: Dominant systems rely heavily on keyword matching or vector based semantic similarity scores (e.g. Cosine Similarity).

Why current approaches are insufficient: As noted by Rokade et al. (2018), these methods often fail to recognize correct answers expressed through non traditional vocabulary or synonyms. Conversely, they may incorrectly award high marks to "hallucinated" answers that contain the right keywords but lack logical reasoning or conceptual depth. [1]

Research Gap: There is a requirement for reasoning aware evaluation methods that prioritize the interpretation of conceptual meaning and logical "flow" over simple pattern matching.

1.3.4 The "Black Box" Problem - Lack of Explainable Feedback

What is missing: Existing AI grading systems typically output a numerical score without providing the underlying rationale or pedagogical justification. [6]

Why current approaches are insufficient: The absence of justification creates significant concerns regarding transparency, fairness, and institutional trust, a sentiment echoed in the studies by Ragolane and Patel (2024). Without "Human in the loop" clarity, instructors cannot verify the AI's decision making process, and students cannot identify their specific learning gaps. [4]

Research Gap: A demand exists for Explainable AI (XAI) in grading that provides descriptive feedback, allowing stakeholders to understand exactly how a specific score was derived based on the marking rubric and identifying specific student misconceptions.

1.3.5 Disconnect from Real World Examination Environments

What is missing: Current systems are largely confined to controlled research environments or specialized online learning platforms (MOOCs). [5]

Why current approaches are insufficient: They are not built to integrate into the established workflows of traditional universities, which involve the physical collection, distribution, and archival of paper based scripts. They lack the "practicality" required for high stakes summative assessment.

Research Gap: There is a need for practical, deployable AI assessment systems that can be applied directly in real world university settings, transitioning seamlessly from physical paper to intelligent digital evaluation.

Justification of Novelty and Creativity

The proposed Unified AI Evaluation Engine introduces a creative paradigm shift by moving from "isolated scoring tools" to an "integrated intelligence framework." Its novelty lies in the synthesis of OCR technology, reasoning focused NLP, and Explainable AI to solve the "Practicality Gap." By focusing on logical path validation rather than simple similarity, and by providing justifiable feedback reports, this project provides a robust, transparent solution specifically tailored for the complexities of modern higher education.

Evidence Based Argument

The necessity for this research is supported by Najafi (2024), whose study highlighted that AI systems currently exhibit an 8% bias against non native speakers due to a focus on linguistic perfection over conceptual understanding. [6] Furthermore, Wang et al. (2025) explicitly identify "Integration of Explainable AI" as the vital future direction for the field. This project directly addresses these evidence based concerns by building a system that prioritizes reasoning, fairness, and transparency. [7]

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate an AI-based automated examination evaluation system by December 2026 that can process handwritten and typed student answers using OCR and Natural Language Processing, provide automated grading with explainable feedback, and achieve at least 85% agreement with human evaluators while reducing the average grading time by 15% in a controlled testing environment**.**

2.1.1 SMART Analysis of the Main Objective

Specific (S) : The objective clearly defines the development of an AI based automated examination evaluation system that processes handwritten and typed student answers. The system will use Optical Character Recognition (OCR) to convert handwritten responses into machine readable text and Natural Language Processing (NLP) to evaluate the content of the answers. The system will also generate explainable feedback to improve transparency in grading.

Measurable (M) : The success of the system can be determined based on the following measurable criteria: Achieving at least 85% agreement with human evaluators in grading results. Reducing the grading time on average by 15% compared to traditional grading. Providing reports that give feedback on grading decisions.

Achievable (A) : The objective is achievable within the scope of a final year research project because the system will integrate existing AI technologies such as OCR and NLP models rather than requiring the development of entirely new algorithms. The project will focus on designing a functional prototype system and evaluating its performance using controlled test datasets.

Relevant (R) : The objective directly addresses the research gaps identified in the literature review, particularly the lack of systems capable of processing handwritten examination scripts and providing explainable AI based grading. The proposed system aims to improve grading efficiency, fairness, and transparency in academic evaluation.

Time bound (T) : The project will be completed within 12 months, following the standard academic research timeline that includes system design, implementation, testing, evaluation, and final documentation.

2.2 Specific Objectives

Objective 1: Develop a Handwritten Answer Digitization Module

Specific: Design and implement an OCR-based preprocessing module that converts handwritten examination scripts into machine-readable text.

Measurable: Evaluate OCR accuracy by measuring text extraction accuracy on a sample dataset of handwritten answers.

Achievable: Existing OCR frameworks and image preprocessing techniques can be integrated and fine-tuned for this purpose.

Relevant: This objective addresses the research gap related to the lack of systems capable of processing handwritten examination scripts.

Time bound: The OCR module will be developed and tested during the initial system implementation phase of the project.

Objective 2: Implement an AI-Based Answer Evaluation Model

Specific: Develop an NLP based grading model capable of evaluating student answers based on semantic meaning rather than simple keyword matching.

Measurable: System performance will be measured by comparing grading results with human evaluators, targeting at least 85% agreement.

Achievable: Pre trained NLP models and transformer based language models can be adapted for answer evaluation tasks.

Relevant: This objective addresses the limitations of traditional keyword based automated grading systems.

Time bound: The grading model will be implemented and evaluated during the system development phase.

Objective 3: Design a Reasoning-Aware Grading Mechanism

Specific: Implement a grading mechanism capable of analyzing logical reasoning and conceptual correctness within student answers.

Measurable: The system will be evaluated based on its ability to correctly identify conceptual answers and detect common misconceptions.

Achievable: Semantic similarity models and reasoning based evaluation methods can be integrated into the grading framework.

Relevant: This objective improves the quality of automated grading by focusing on conceptual understanding rather than surface-level text matching.

Time bound: This component will be developed after the initial NLP grading model is implemented.

Objective 4: Develop an Explainable AI Feedback System

Specific: Design a feedback module that generates clear explanations showing how grades are assigned.

Measurable: The system will produce feedback reports linked to grading criteria and evaluation outcomes.

Achievable: Explainable AI techniques can be integrated with the grading model to generate interpretable feedback.

Relevant: This addresses the lack of transparency in existing AI grading systems.

Time bound: The explainable feedback module will be developed during the later stages of system implementation.

Objective 5: Integrate System Components into a Unified Evaluation Platform

Specific: Develop a web based platform that integrates OCR processing, AI grading, and feedback generation.

Measurable: The platform will allow instructors to upload answer scripts and receive automated evaluation results.

Achievable: Existing web technologies and AI frameworks can be integrated to build the platform.

Relevant: This objective ensures that the proposed solution is practical and usable in real academic environments.

Time bound: The integration will be completed during the system development phase before testing.

Objective 6: Evaluate System Performance and Reliability

Specific: Conduct performance testing of the proposed system using sample examination datasets.

Measurable: Evaluation metrics include grading accuracy, 15% reduction in grading time, and inter rater reliability ≥ 0.85.

Achievable: Comparative testing with human examiners can be conducted using collected datasets.

Relevant: This ensures the effectiveness and reliability of the system for real world academic use.

Time bound: Evaluation and validation will be conducted during the final testing phase of the research.

3. METHODOLOGY

3.1 Research Approach

This research adopts a Design Science Research (DSR) approach, which focuses on the development and evaluation of innovative technological artifacts to solve real world problems. The DSR methodology is appropriate for this study because the primary objective is to design and implement a functional system the Unified AI Evaluation Engine to address limitations in traditional examination grading processes.

The DSR process for this project follows several stages: problem identification, system design, artifact development, and evaluation. Initially, the limitations of manual and existing automated grading systems are analyzed through literature review and problem analysis. Based on these findings, the proposed system architecture is designed and implemented as a prototype. The final stage involves evaluating the system using controlled experiments and comparing its grading performance with human examiners.

In addition to the artifact development process, an experimental evaluation approach will be used to measure the effectiveness of the system. The automated grading results produced by the system will be compared with grades assigned by human evaluators to assess accuracy, consistency, and efficiency.

3.2 Technical Methods and Algorithms

In order to assist the computer system in the automated assessment of the student's examination script answers, the proposed system will incorporate various artificial intelligence techniques and processing modules.

OCR and Handwritten Script Digitization

The proposed system will use Optical Character Recognition (OCR) technology to digitize the handwritten examination script. The system will use tools such as Tesseract OCR or Amazon Textract to convert the scanned handwritten script into digital text. The system will also use image processing techniques to enhance the accuracy of the script. The spatial data of the original script will be maintained so that the structure of the original script will be preserved.

Knowledge Fusion and Semantic Representation

The proposed system will also use knowledge fusion techniques to represent the conceptual relationships between the concepts presented in the student's script. The proposed system will also use graph based representations using Graph Neural Networks (GNNs) to represent the relationships between the concepts presented in the student's script. This will allow the system to link the important concepts presented in the script.

Reasoning Aware Grading Engine

The grading engine will be based on a Large Language Model (LLM), which will be fine tuned to perform answer evaluation tasks. For this purpose, an LLM such as Llama 3 or Mistral can be fine tuned using a Chain of Thought (CoT) prompt. This will enable the grading engine to analyze the logical

reasoning provided by the student in the answer. Such a grading engine will be able to validate logical paths in student responses.

Explainable AI Feedback Module

To overcome the transparency problem associated with automated grading systems, Explainable AI (XAI) techniques will be incorporated into the proposed system. SHAP (SHapley Additive Explanations) or LIME (Local Interpretable Model agnostic Explanations) techniques will be used to identify the portion of the student's answer that contributed to the overall grade.

3.3 Tools and Platforms

The tools and platforms necessary in the development and deployment of the proposed system are discussed in this section.

Programming Language

The programming language necessary in the development of the proposed system is Python, based on its support in machine learning, NLP, and image processing.

AI and NLP Libraries

The proposed system can be developed with the support of spaCy, NLTK, and Hugging Face Transformers libraries, which can be used in the development of Natural Language Processing. Moreover, deep learning frameworks such as PyTorch can be used in the development of the proposed system.

Backend Framework

The proposed system can be developed with the support of FastAPI, which can be used in handling multiple scripts at the same time, especially in an examination environment.

Database

The database that will be used in storing the data related to the examination, evaluation results, and output from the system will be PostgreSQL or something similar. The pgvector extension will be utilized in storing the vector data that is produced from the NLP models.

Deployment Environment

The system will be deployed using Docker containerization technology. Docker provides the application the ability to be deployed on any computing environment. Additionally, it improves the scalability of the system and the ease of deployment.

3.4 Data Collection and Ethical Considerations

Data Collection

The system will be trained and tested using historical examination data collected from various university sources. This will include scanned handwritten answer scripts and possibly some examples of student responses and rubrics that an instructor might use. This will be instrumental in testing the automated grading system.

Ethical Considerations

During the research and testing of the system, various ethical issues will be considered.

First and foremost, the research will consider the anonymization of the data collected. This will be instrumental in ensuring that any identifiable features of the students, such as their names or identification numbers, are removed from the examination scripts collected for the purpose of the research.

Secondly, the research will consider the elimination of algorithmic bias. A wide range of data will be collected and tested using the system. This is an important aspect that was considered in the research conducted previously on the topic.

Lastly, if the research requires the collection of data directly from the subjects, their consent will be sought before the research begins.

3.5 Validation Strategy and Performance Metrics

The proposed system will be subjected to a Human in the Loop (HITL) validation method. Here, the automated grading system will grade a set of examination answers. Meanwhile, a set of human examiners will grade the same set of examination answers according to the official marking scheme. The graded results will be compared to evaluate the accuracy and reliability of the proposed system.

The performance of the proposed system will be evaluated based on the following metrics.

Inter Rater Reliability (IRR)

The agreement between the proposed system and the human evaluators will be assessed by using statistical measures such as Cohen's Kappa and the Pearson Correlation Coefficient. The reliability of the proposed system will be IRR ≥ 0.85. This implies a strong agreement between the proposed system and the human evaluators. [8]

Misconception Detection Accuracy

The accuracy of the proposed system in identifying conceptual errors will be assessed by using the Precision and Recall metrics. These metrics will evaluate the accuracy of the proposed system in identifying misconceptions.

Efficiency Metrics

The time taken by the system for grading will be compared with the time taken by the manual grading method. The objective of the research is to achieve a reduction of at least 15% in the time taken for grading without compromising the accuracy of the grading results.

Bias Evaluation

To evaluate the fairness of the grading results, the system will be employed to evaluate the bias of the grading results by comparing the scores of students from different linguistic groups.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

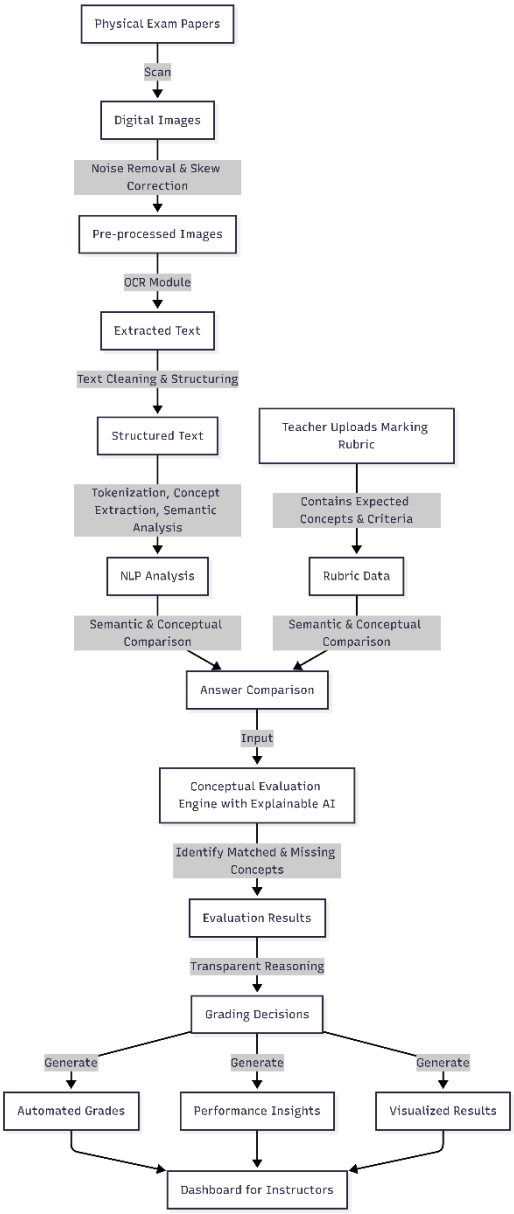


Figure 1: High level system diagram

The high level system architecture of the proposed module focuses on converting physical exam answer scripts into structured data for AI based evaluation. Users upload images of handwritten answer papers through a web or mobile interface. The images are first processed through an image preprocessing layer to improve quality and readability. An OCR engine then extracts handwritten text while preserving the structure of the answers. The extracted content is passed to the AI analysis layer, where semantic models analyze the responses and identify key concepts or misconceptions. Finally, the processed data is sent to the central evaluation system to support automated grading and feedback generation.

5. USER REQUIREMENTS

This section outlines the specific needs of the various stakeholders and defines the functional and non-functional capabilities of the Unified AI Evaluation Engine.

5.1 Stakeholder Identification

The success of the evaluation system depends on addressing the needs of four primary stakeholder groups:

Primary Users (Academic Staff/Examiners): Instructors and professors who define marking rubrics, oversee the automated grading process, and validate the AI-generated results.

Secondary Users (Students): Individuals receiving the grades and feedback. Their primary need is fairness, transparency, and timely results.

Tertiary Users (University Administrators): Institutional leaders who require high-level analytics on performance trends, system reliability, and reduction in administrative overhead.

Technical Stakeholders (System Developers/IT Staff): Responsible for the maintenance, integration, and security of the AI infrastructure within the university's existing Learning Management Systems (LMS).

5.2 Functional Requirements (FR)

Functional requirements define the specific actions the system must perform to fulfill the research objectives.

FR1: Multi Modal Data Ingestion

The system shall ingest and process multiple data formats, including digitized handwritten answer scripts (via OCR), structured diagram data, and transcribed oral viva audio files.

FR2: Reasoning Aware Grading Engine

The system shall evaluate student responses based on logical path validation and conceptual coverage rather than simple keyword matching, utilizing fine-tuned Large Language Models (LLMs).

FR3: Misconception Identification

The system shall detect and flag specific conceptual errors or logical gaps in student responses, providing a diagnostic summary of where the student’s understanding deviates from the marking scheme.

FR4: Automated Feedback Generation

The system shall generate a human readable "Evaluation Justification Report" for every graded script, explaining the rationale behind the awarded marks based on the provided rubric.

FR5: Human in the Loop Validation Interface

The system shall provide a dashboard for examiners to review, override, or confirm AI generated grades and feedback before final publication.

FR6: Cross Modal Knowledge Verification

The system shall perform consistency checks between different modalities (e.g., verifying if the logic in a student's diagram aligns with their oral explanation in the viva).

FR7: User Authentication and Access Control

The system shall allow authorized users (examiners and administrators) to securely log in and access system features based on their assigned roles.

5.3 Non Functional Requirements (NFR)

Non functional requirements define the quality attributes and constraints of the system.

NFR1: Performance (Efficiency)

The engine shall process and grade a standard 5 page handwritten examination script in under 60 seconds, aiming for a total grading time reduction of 15% compared to manual evaluation.

NFR2: Accuracy and Reliability

The system shall maintain an Inter Rater Reliability (IRR) score of >= 0.85 when compared against a panel of human expert examiners.

NFR3: Fairness and Bias Mitigation

The system shall be audited to ensure that grading variance between different demographic groups (e.g., native vs. non native speakers) is minimized, addressing the 8% bias gap identified in existing literature. [6]

NFR4: Usability

The examiner dashboard shall adhere to WCAG 2.1 accessibility standards and provide an intuitive UI/UX, ensuring that instructors can review a script with minimal technical training.

NFR5: Security and Data Privacy

The system shall ensure end-to-end encryption for all student data and examination scripts. It must strictly comply with data protection regulations (e.g., GDPR or local university policies) by anonymizing Personally Identifiable Information (PII) during the AI processing phase.

NFR6: Scalability

The system architecture shall be containerized using Docker, allowing it to scale horizontally to handle high concurrency demands during peak end of semester examination periods.

6. COMMERCIALIZATION PLAN

The target market for the proposed system includes universities, higher educational institutions, and examination boards that deal with high volumes of student assessment. These are the main target markets, and they are often faced with problems concerning the time spent on grading and ensuring that evaluation is consistent. The secondary target market could include online learning platforms, certification, and training centers. The ideal customer for the proposed system could be a lecturer or an examiner in a university whose main task is to grade numerous answer scripts. These customers need assistance in reducing their workload and increasing their efficiency in the process.

The value proposition of the proposed system is that it enhances the efficiency, accuracy, and transparency of academic evaluation. The system uses Optical Character Recognition (OCR) and Natural Language Processing (NLP) technologies, enabling it to evaluate and analyze handwriting in answer scripts. This helps in reducing the time spent on grading, and the evaluation is more consistent. The system also helps in providing transparent feedback on how the evaluation is done.

This can be achieved by adopting a Software as a Service (SaaS) business model, whereby institutions can purchase the software with monthly or yearly subscription fees. Institutions can be offered various subscription fees depending on the size of the institution and the number of exam scripts processed. Additional revenue can be generated by licensing the grading engine to ed tech companies or API integration with existing Learning Management Systems.

Cost estimation comprises development costs and operational costs. Development costs include the development of AI models, software development, and user interface development. Operational costs include cloud costs, data storage, and maintenance costs. Additional costs may include marketing costs, maintenance costs, and periodic update costs to improve the performance and reliability of the software.

The pricing strategy will be based on a tiered subscription model to accommodate the varying needs and sizes of the institutions. The pricing will be such that smaller institutions can opt for basic subscription plans, while larger universities may opt for enterprise plans with higher processing capacity. Another pricing model that can be considered is a pay per script model, wherein institutions will be charged based on the number of exam papers processed by the system.

The proposed system will have a number of competitive advantages over other existing automated grading tools. First, the proposed system will be able to process handwritten scripts, whereas most existing automated tools are limited to typed answers. Second, the proposed system will be based on reasoning based evaluation, not just keyword matching. Third, the use of explainable AI will enhance the trust factor for automated grading tools.

Another important aspect to be considered for the proposed system is intellectual property. The proposed system can be protected under copyright law for the software. Additionally, new AI grading approaches or system components can be protected under patent law. Furthermore, the data and evaluation approaches can be considered proprietary, provided the data privacy laws are followed within educational institutions.

7. BUDGET AND JUSTIFICATION

Table 2: Budget justification

Item	Technology / Resource	Estimated Cost (LKR)
Software and Services		
1	NLP Model (Hugging Face NLP)	62,000.00
2	OCR Model (Google Cloud Vision)	20,000.00
3	Cloud Storage + Network (Microsoft Azure)	30,000.00
4	Cloud Hosting (Microsoft Azure)	55,000.00
5	Wi-Fi / broadband connection (8 months)	19,200.00
Data Collection		
6	Data Collection Materials & Documentation	3000.00
Contingency & Miscellaneous		
7	Unexpected Expenses (10% buffer)	18,900.00
Total		69,520.00

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task breakdown

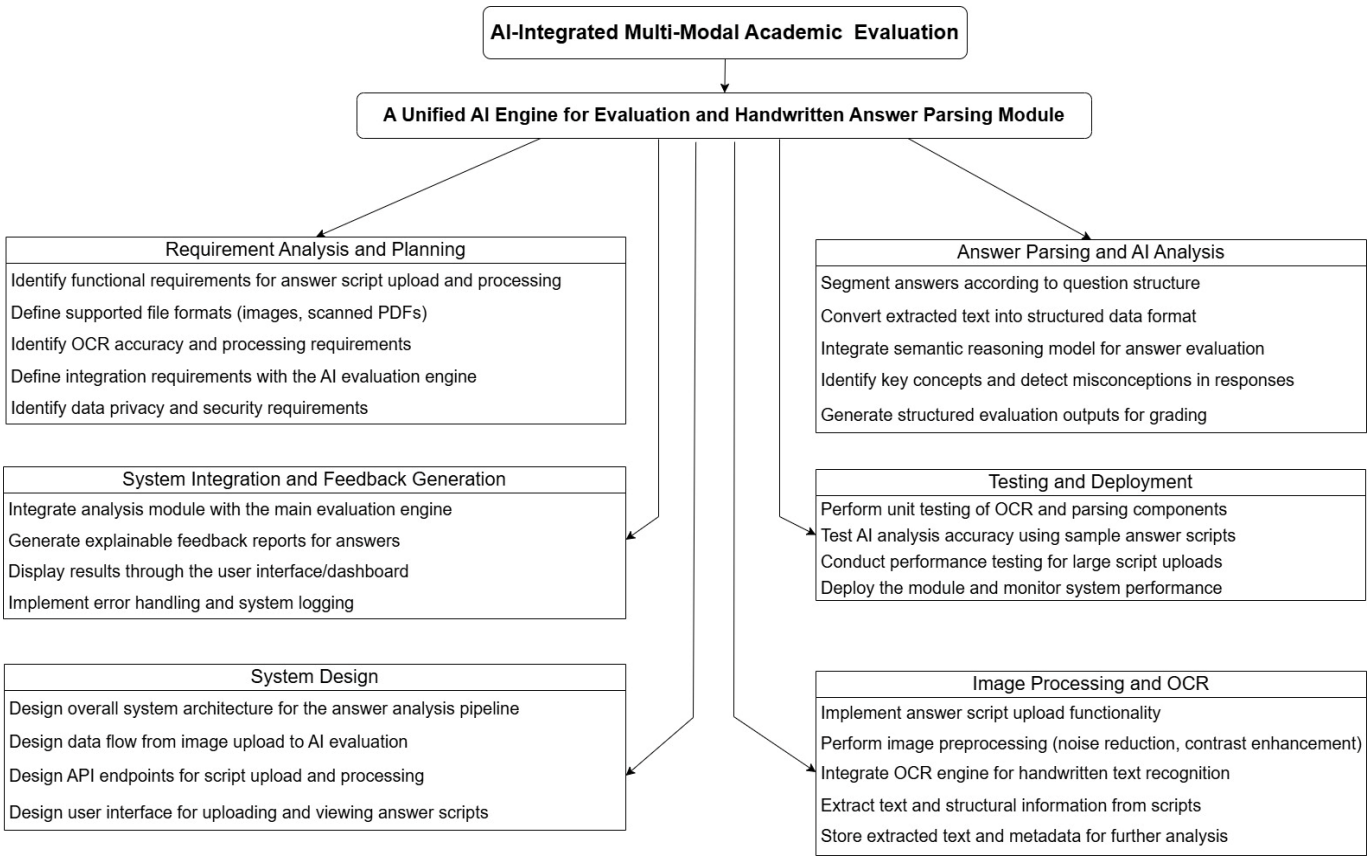


Figure 2: Work Breakdown Structure

8.2 Timeline Alignment

Table 3: Timeline Alignment

WBS Component	Subtasks	Timeline (Months)	Notes
1. Requirement Analysis and Planning	1.1–1.5	Month 1–2	Understand functional, OCR, AI integration, and security requirements. Conduct stakeholder discussions and finalize specs.
2. System Design	2.1–2.4	Month 2–3	Design overall architecture, data flow, API endpoints, and UI wireframes. Approve design documents.
3. Image Processing and OCR	3.1–3.5	Month 3–6	Implement upload functionality, preprocessing, OCR integration, text extraction, and metadata storage. Iterate to improve recognition accuracy.
4. Answer Parsing and AI Analysis	4.1–4.5	Month 5–8	Segment answers, convert to structured data, integrate semantic reasoning models, detect misconceptions, and prepare evaluation outputs.
5. System Integration and Feedback Generation	5.1–5.4	Month 7–10	Integrate parsing module with main AI engine, generate feedback reports, connect UI/dashboard, and implement logging/error handling.
6. Testing and Deployment	6.1–6.4	Month 10–12	Unit testing of OCR/parsing, AI accuracy testing, performance testing for large scripts, deployment, and monitoring.

8.3 Balanced workload distribution

Table 4: Balanced workload distribution

Member	Individual Contribution	Shared Workload	Balance Justification
Member 1	Diagram extraction, AI-based answer evaluation, semantic reasoning integration	Proposal, dataset preparation, testing, final report	Handles tasks focused on structural understanding of answers and evaluation logic. Balanced with analytics and OCR work by others.
Member 2	Analytics module, feedback report generation, dashboard display	Proposal, dataset preparation, testing, final report	Responsible for transforming parsed answers into actionable insights. Complements Member 1's evaluation by focusing on presentation and metrics.
Member 3	Viva evaluation integration, grading logic for viva, report consolidation	Proposal, dataset preparation, testing, final report	Focuses on oral/written assessment integration, which is lighter than continuous parsing, but workload is balanced by involvement in shared tasks.
Member 4	Handwritten text extraction, OCR preprocessing, text-to-structured-data conversion, error handling	Proposal, dataset preparation, testing, final report	Handles heavy technical extraction work, balanced by distributing testing and reporting tasks among all members.

9. GANTT CHART

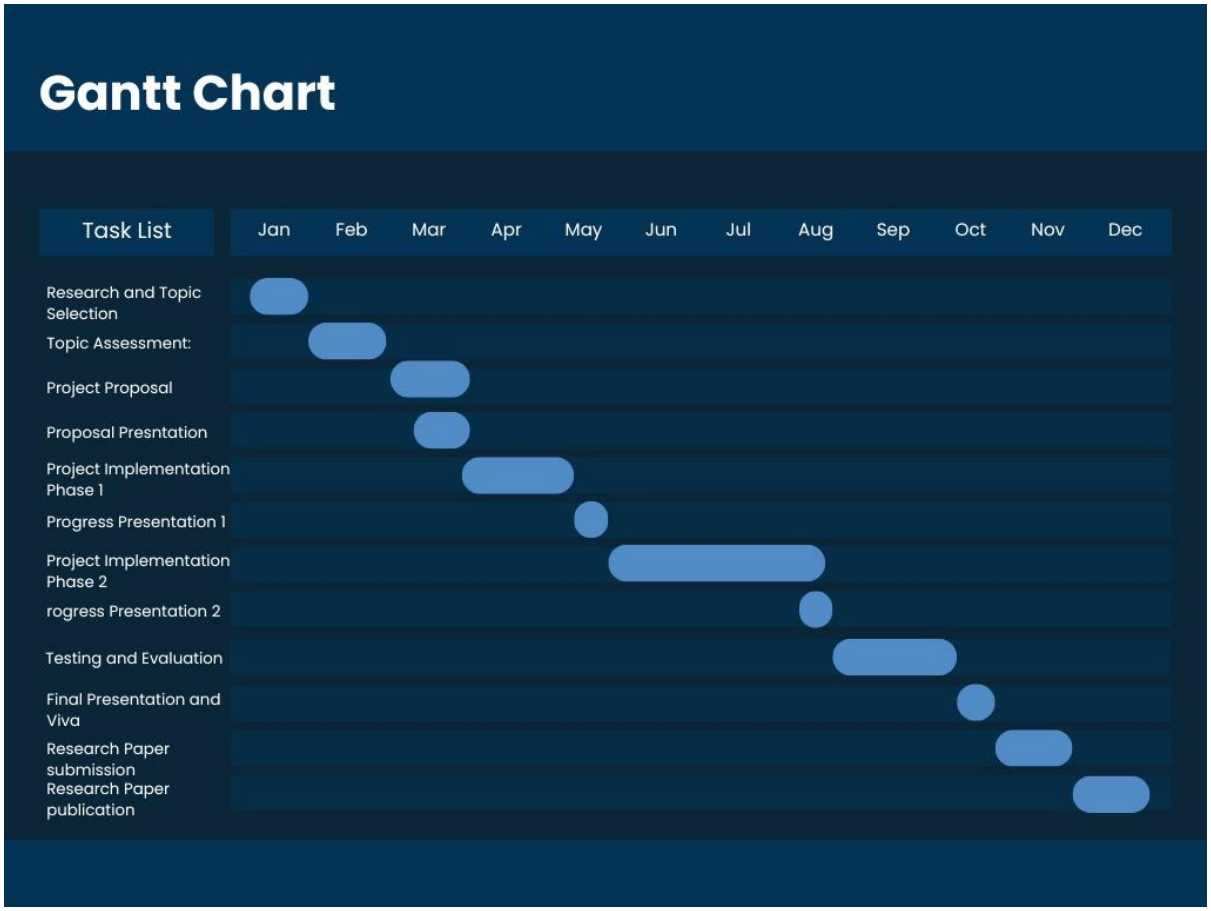


Figure 3: Gantt chart

10.REFERENCES LIST

[1] A. Rokade, B. Patil, S. Rajani, S. Revandkar and R. Shedge, "Automated Grading System Using Natural Language Processing," IEEE, 2018.

[2] A. G. . F. T. M. M. . T. Z. . M. Z. A. W. H. and . S. S. , "BEYOND HUMAN SUBJECTIVITY AND ERROR: A NOVEL AI GRADING SYSTEM," IU International University of Applied Sciences, 2024.

[3] V. Suresh, R. Agasthiya and J. Ajay, "AI based Automated Essay Grading System using NLP," IEEE, 2023.

[4] D. S. P. Ragolane Mahlatse, "Artificial Intelligence, Digital Grades and Human Voices: Students," International J. Soc. Sci. & Education, 2024.

[5] H. Aldriye, A. Alkhalaf and M. Alkhalaf, "Automated Grading Systems for Programming Assignments," International Journal of Advanced Computer Science and Applications, 2019.

[6] V. R. P. U. o. F. R. S. P. S. U. H. T. C. C. Fazil T. Najafi University of Florida, "AI-Assisted Grading – A Study on Efficiency and Fairness," Mississippi State University, Mississippi, 2025.

[7] J. Wang, Y. Hussain, X. Zhang and C. Mao, "AI-Based Automated Grading Systems: Opportunities, Challenges, and Future Directions," Junyao Wang, 2025.

[8] A. D. K. ,. Z. V. S. ,. a. J. M. J. Bojan B. Tomic, "An AI-based Approach for Grading," IEEE, 2023.

[9] S. H. M. F. Z. Z. C. Z. K. Tiffany Wenting Li, " Effects of AI Grading Mistakes on Learning," 2023.

[10] K. Matthews and T. L. H. Janicki, "Implementation of an Automated Grading System with an Adaptive Learning Component to Affect Student Feedback and Response Time," Journal of Information Systems Education, 2019.

11.APPENDICES